

Effect of Pose-Diverse Training on Cross-Pose Face Recognition and Learned CNN Representations

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Abstract— This study investigates whether pose-diverse training improves cross-pose face recognition and learned CNN representation stability. Two CNN classifiers with identical architectures and training budgets were evaluated on 153 identities from the Multi-PIE dataset. Model A was trained using frontal images only, whereas Model B used 0° , $\pm 30^\circ$, and $\pm 60^\circ$ yaw images. Across three random seeds, Model A achieved $19.17 \pm 1.33\%$ test accuracy, compared with $86.45 \pm 0.51\%$ for Model B. Embedding analysis also showed that Model B preserved substantially higher same-identity similarity across pose variation. However, performance declined at extreme unseen yaw angles, indicating that pose-diverse training improves cross-pose robustness but does not fully eliminate extreme-pose sensitivity.

1. INTRODUCTION

Head pose variation is a major source of appearance change in face recognition and has long been recognized as a challenging factor for recognition across viewpoints [1]. As the head rotates away from the frontal view, facial structures become displaced, partially occluded, or no longer visible. A convolutional neural network (CNN) trained primarily on frontal faces may therefore learn identity representations that work well for frontal inputs but become less reliable when the same person is observed from a different yaw angle. Deep learning-based face recognition generally relies on hierarchical representations that encode identity-related information from facial images [2].

One possible way to improve robustness is to expose the model to multiple head poses during training. Instead of learning identity from a single frontal appearance, pose-diverse training allows the CNN to observe how the appearance of the same identity changes across viewpoints. However, it is important to determine whether this additional pose diversity improves only recognition at poses already observed during training or whether it also helps the learned representation generalize to previously unseen yaw angles.

This study investigates this question through a controlled comparison between two CNN classifiers with the same architecture, optimization settings, and number of training images. **Model A** is trained using frontal images only, while **Model B** is trained using images at 0° , $\pm 30^\circ$, and $\pm 60^\circ$ yaw. Both models are evaluated on the same held-out test set covering yaw angles from 0° to 90° . In addition to classification accuracy and Macro-F1, the study examines the cosine similarity between non-frontal embeddings and their corresponding frontal embeddings to analyze pose-induced changes in the learned representation.

Research Question: *Does pose-diverse training improve cross-pose CNN face recognition and reduce pose-induced representation shift compared with frontal-only training?*

Two hypotheses are evaluated. **H1:** recognition performance of the frontal-only model will decrease as the absolute yaw angle increases, whereas pose-diverse training will provide substantially better non-frontal recognition. **H2:** same-identity embedding similarity to the frontal representation will decrease with increasing yaw, but the pose-diverse model will preserve higher similarity than the frontal-only model.

2. RELATED CONCEPT

Convolutional neural networks learn hierarchical image representations through repeated convolutional and nonlinear transformations [3]. For an input face image x , the feature extractor can be represented as

$$z = f_\theta(x),$$

where z is a learned feature embedding and f_θ denotes the CNN parameterized by θ . Learned feature embeddings are widely used in modern face-recognition systems to represent facial identity in a compact feature space [4], [5]. The final classifier then predicts the identity from this representation. Ideally, images of the same person should produce similar embeddings even when nuisance factors such as illumination or head pose change.

Head rotation, however, changes the visible facial structure. As yaw increases, facial components shift spatially and some regions become partially or fully occluded. Consequently, two images of the same identity observed at different yaw angles may produce substantially different learned representations. In this study, this phenomenon is referred to as **pose-induced representation shift**.

To quantify this effect, the similarity between the embedding of a non-frontal image and the corresponding frontal embedding of the same identity is measured using cosine similarity:

$$S_i(\phi) = \frac{z_i^{0^\circ} \cdot z_i^\phi}{\|z_i^{0^\circ}\| \|z_i^\phi\|}, \quad (1)$$

where $z_i^{0^\circ}$ is the frontal embedding of identity i , and z_i^ϕ is the embedding at yaw angle ϕ . A higher value indicates that the learned identity representation remains more similar to the frontal representation despite pose variation.

Pose-diverse training is expected to reduce this representation shift by exposing the CNN to multiple appearances of the same identity during training. In addition to performance at observed training poses, the experiment also evaluates whether this learned robustness extends to **unseen intermediate poses** and to **extreme poses outside the training range**.

3. DATASET AND PREPROCESSING

The experiments were conducted using a filtered subset of the Multi-PIE face dataset [6]. The final dataset contained **153 identities**, with images captured under multiple head poses and illumination conditions. To reduce an additional appearance-related nuisance factor, images and identities containing eyeglasses were excluded from the experimental subset. Each identity contained **13 signed yaw conditions**, corresponding to 0° , $\pm 15^\circ$, $\pm 30^\circ$, $\pm 45^\circ$, $\pm 60^\circ$, $\pm 75^\circ$, and $\pm 90^\circ$, with **20 illumination conditions** per pose. Dataset auditing confirmed a complete set of **39,780 images** with no missing pose–illumination combinations or duplicate samples.

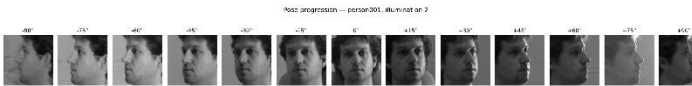


Fig. 1. Example images of the same identity across increasing head-yaw angles in the Multi-PIE subset used in this study.

The prediction target was **person identity**, resulting in a 153-class closed-set identification problem. The same identities were retained across the training, validation, and test partitions; however, the partitions were separated by **illumination condition** to prevent the same illumination instance from appearing across splits. Twelve illumination conditions were assigned to training, four to validation, and four to testing. The final common test set contained **7,956 images** covering all 13 yaw conditions. The same fixed data split was used for all compared models and repeated-seed experiments.

All original images were 128×128 RGB images. Preprocessing converted each image to grayscale, preserved the spatial resolution at 128×128 , converted pixel values to floating point, and normalized the intensity range to $[0,1]$.

During training only, mild brightness and contrast augmentation of approximately $\pm 10\%$ was applied.

Geometric augmentation, including horizontal flipping, rotation, translation, perspective transformation, and synthetic yaw generation, was intentionally excluded because head pose was the primary experimental variable. No augmentation was applied to validation or test images.

4. METHODS

A controlled experiment was designed to isolate the effect of training-pose diversity on cross-pose face recognition. Two CNN classifiers were trained using the same architecture, optimization settings, data split, and number of training images. The primary experimental variable was the distribution of head-yaw angles included during training. The overall experimental workflow is illustrated in Fig. 2.

4.1 Baseline and Experimental Comparison

Model A served as the frontal-only baseline. It was trained exclusively using images at 0° yaw, with 12 training images per identity. **Model B** used the same total number of original training images but distributed them across multiple head poses: 0° , $\pm 30^\circ$, and $\pm 60^\circ$. For each identity, Model B used four frontal images and two images from each non-frontal training pose, resulting in the same total training budget of **1,836 images** for both models.

This design ensured that any performance difference between the two models could be primarily attributed to **training-pose diversity rather than the number of training samples or model capacity**.

4.2 Controlled Experimental Design

The independent variable was the **training-pose distribution**, while the CNN architecture, preprocessing procedure, training-image budget, optimizer, learning rate, batch size, number of epochs, validation set, and test set were held constant. Both models were trained and evaluated using the same identity set and the same fixed illumination-based data partitions.

Validation used held-out illumination conditions at 0° , $\pm 30^\circ$, and $\pm 60^\circ$. The final test set was shared by both models and contained all 13 signed yaw conditions from -90° to $+90^\circ$. Test data were not used for model selection or hyperparameter adjustment.

To evaluate run-to-run variability, the complete training experiment was repeated using three random seeds: **42**, **123**, and **2026**. The dataset split remained fixed across all runs; only stochastic training factors such as parameter initialization, data shuffling, and augmentation varied with the seed.

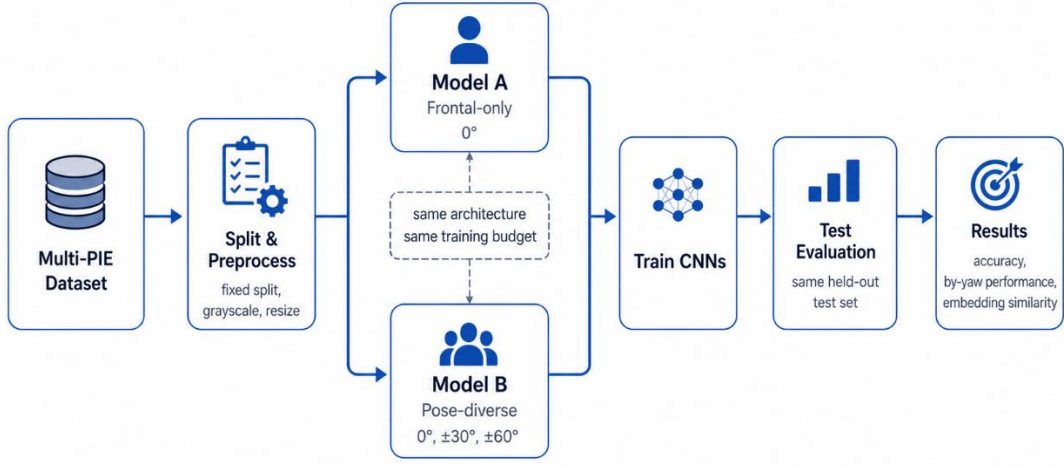


Fig. 2. Overview of the controlled experimental pipeline comparing frontal-only and pose-diverse CNN training.

4.3 CNN Architecture

Both models used the same CNN architecture. Four convolutional stages progressively increased the number of feature channels while reducing spatial resolution. The final convolutional representation was pooled to 4×4 , projected to a 128-dimensional embedding, and classified into 153 identities.

4.4 Training Protocol

The models were implemented in PyTorch and trained using GPU acceleration. Cross-entropy loss was optimized using Adam optimizer [7] with a learning rate of 3×10^{-4} , weight decay of 1×10^{-5} , and a mini-batch size of 32. Each model was trained for a fixed budget of **40 epochs**. Gradient clipping with a maximum norm of 5.0 was used to improve numerical stability.

Mild brightness and contrast augmentation was applied only to training images. No geometric augmentation was used. The best checkpoint from each run was selected according to the highest validation Macro-F1 score; when two checkpoints obtained the same Macro-F1, the checkpoint with the lower validation loss was selected.

TABLE I. CNN ARCHITECTURE USED FOR BOTH MODELS

Stage	Configuration
Block 1	Conv 3×3 , 32×2 + BN + ReLU + MaxPool
Block 2	Conv 3×3 , 64×2 + BN + ReLU + MaxPool
Block 3	Conv 3×3 , 128×2 + BN + ReLU + MaxPool
Block 4	Conv 3×3 , 256×2 + BN + ReLU + MaxPool
Spatial pooling	Adaptive average pooling 4×4
Fully connected	$4096 \rightarrow 256$
Embedding	128-D
Classifier	$128 \rightarrow 153$ classes

4.5 Evaluation Metrics

Classification performance was evaluated using **top-1 accuracy** and **Macro-F1**. Performance was reported both overall and separately by absolute yaw angle,

$$|\phi| \in \{0^\circ, 15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ\}.$$

For Model B, test poses were additionally grouped into three categories: **seen training poses** ($0^\circ, \pm 30^\circ, \pm 60^\circ$), **unseen interpolation poses** ($\pm 15^\circ, \pm 45^\circ$), and **extreme extrapolation poses** ($\pm 75^\circ, \pm 90^\circ$).

Representation robustness was evaluated using the cosine similarity defined in Eq. (1) between each non-frontal embedding and the frontal embedding of the same identity under the same held-out illumination condition. All primary quantitative results are reported as

$$\boxed{\text{mean} \pm \text{standard deviation}}$$

across the three training seeds.

5. RESULTS

Final test performance was aggregated across three independent training runs using random seeds 42, 123, and 2026. Unless otherwise stated, quantitative results are reported as **mean $\pm$ standard deviation across seeds**. Model A represents frontal-only training, whereas Model B represents pose-diverse training.

5.1 Training Behavior

The representative training curves from seed 42 are shown in Fig. 3. Both models eventually achieved approximately 100% training accuracy, indicating that the CNN had sufficient capacity to fit their respective training sets. However, their validation behavior differed substantially. Model A reached 100% training accuracy within the first few epochs while its validation accuracy remained near 30%, producing a large train-validation gap. In contrast, Model B progressively improved on both the training and validation sets and reached nearly perfect validation performance.

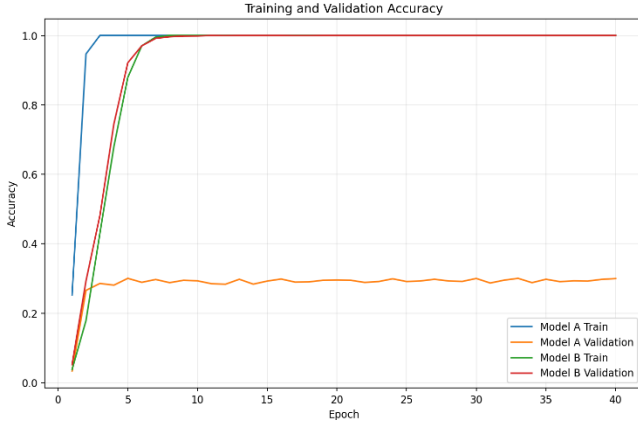


Fig. 3. Representative training and validation accuracy curves for seed 42. Model A rapidly fits the frontal-only training set but exhibits a large validation gap, whereas Model B achieves high training and validation performance.

5.2 Overall and Cross-Pose Recognition

Across the complete test set, Model A achieved an average accuracy of $19.17 \pm 1.33\%$ and a Macro-F1 score of $23.40 \pm 0.88\%$. Model B achieved $86.45 \pm 0.51\%$ accuracy and $87.03 \pm 0.25\%$ Macro-F1. The average improvement in accuracy was therefore 67.28 ± 0.91 percentage points in favor of pose-diverse training.

As shown in Fig. 4, both models achieved 100% recognition at the frontal pose. Their performance diverged rapidly as the yaw angle increased. Model A decreased to $42.57 \pm 2.90\%$ at 15° and below 16% from 30° onward. Model B maintained above 93% accuracy from 15° through 60° , before decreasing to $68.27 \pm 2.90\%$ at 75° and $53.89 \pm 1.82\%$ at 90° .

TABLE II. OVERALL TEST PERFORMANCE ACROSS THREE SEEDS

Model	Accuracy	Macro-F1
Model A — Frontal-only	$19.17 \pm 1.33\%$	$23.40 \pm 0.88\%$
Model B — Pose-diverse	$86.45 \pm 0.51\%$	$87.03 \pm 0.25\%$

5.3 Generalization by Pose Category

Model B showed different behavior across the three pose categories. At **seen training poses** ($0^\circ, \pm 30^\circ, \pm 60^\circ$), its average accuracy was $99.90 \pm 0.03\%$. At **unseen interpolation poses** ($\pm 15^\circ, \pm 45^\circ$), accuracy remained high at $94.99 \pm 0.67\%$. In contrast, performance decreased to $61.08 \pm 1.10\%$ for **extreme extrapolation poses** ($\pm 75^\circ, \pm 90^\circ$). This pattern was consistent across the three random seeds.

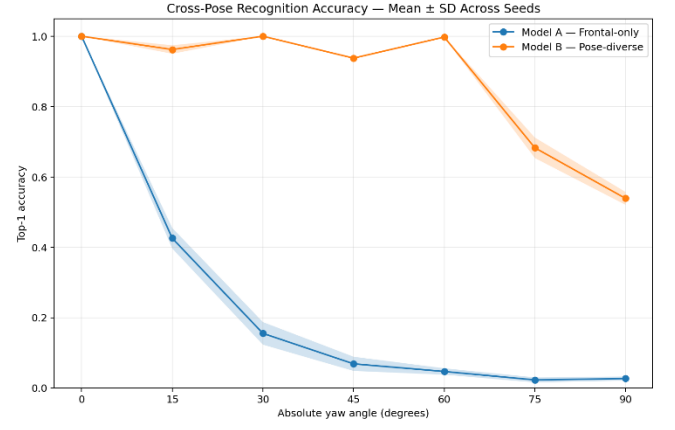


Fig. 4. Cross-pose recognition accuracy across absolute yaw angles. Lines represent the mean across three random seeds, and shaded regions indicate ± 1 standard deviation.

TABLE III. MODEL B PERFORMANCE BY POSE CATEGORY

Pose category	Yaw angles	Accuracy
Seen	$0^\circ, \pm 30^\circ, \pm 60^\circ$	$99.90 \pm 0.03\%$
Unseen interpolation	$\pm 15^\circ, \pm 45^\circ$	$94.99 \pm 0.67\%$
Extreme extrapolation	$\pm 75^\circ, \pm 90^\circ$	$61.08 \pm 1.10\%$

5.4 Representation Similarity

The embedding analysis showed a similar pose-dependent pattern. For Model A, the mean same-identity cosine similarity to the frontal representation decreased sharply from 1.0 at 0° to approximately 0.46 at 15° , 0.29 at 30° , and approximately 0.13 at 90° . Model B consistently preserved substantially higher similarity across non-frontal views. Its mean similarity remained approximately 0.86 at 15° , 0.93 at 30° , 0.81 at 45° , and 0.90 at 60° , before decreasing to approximately 0.64 and 0.59 at 75° and 90° , respectively.

These results show that the classification-performance differences between the two models are accompanied by corresponding differences in the stability of their learned identity representations across pose variation.

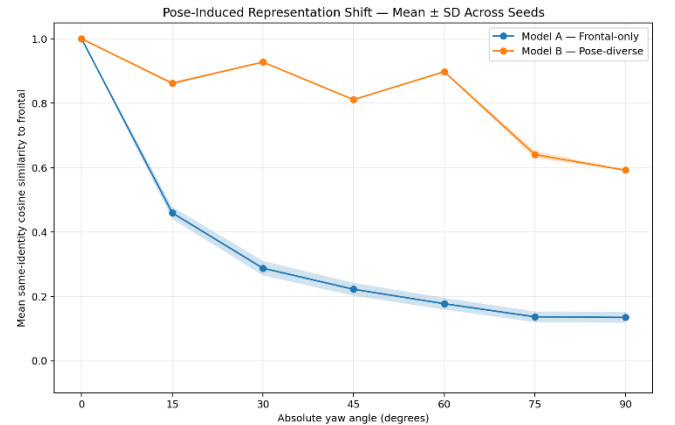


Fig. 5. Same-identity cosine similarity to the frontal embedding across yaw angles (mean $\pm$ SD over three seeds).

6. ERROR ANALYSIS

Error analysis was performed to determine whether the improvement from pose-diverse training was consistent across individual test cases and to identify the remaining failure modes. In the representative seed-42 run, the majority of disagreements favored Model B: 5,296 test images were misclassified by Model A but correctly classified by Model B, whereas only 15 images showed the opposite outcome. Additional cross-seed analysis showed that **4,258 test images were consistently misclassified by Model A and correctly classified by Model B in all three random-seed runs**. In contrast, **556 images were misclassified by both models in all three runs**, and Model B had **570 persistent failures** across all seeds.

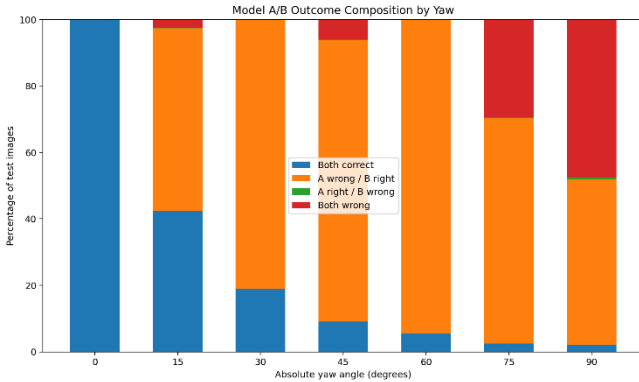


Fig. 6. Distribution of prediction outcomes across absolute yaw angles for the representative seed-42 run. The proportion of cases where Model B corrects Model A increases substantially at non-frontal poses, while joint failures become more frequent at extreme yaw.

The remaining Model B failures were strongly concentrated at extreme yaw angles. Persistent failures were rare or absent at frontal and observed training poses, while the majority occurred at 75° and 90°. This pattern is consistent with the decline in recognition accuracy observed beyond the maximum training yaw of 60°. Representative failure examples also showed that extreme profile views remove or heavily occlude facial regions that are visible in frontal and moderate-pose images.

Error patterns were further examined using aggregate confusion matrices and per-identity recall across the three seeds. Model B showed a strong diagonal structure in the confusion matrix, indicating that most identities were classified consistently, whereas Model A exhibited substantially more off-diagonal confusion. Detailed confusion matrices, persistent-error cases, and per-identity recall distributions are provided in the Appendix.

7. DISCUSSION

The results strongly support the hypothesis that training-pose diversity improves cross-pose face recognition. Although Model A achieved perfect recognition at the frontal pose, its performance degraded rapidly as yaw increased. This indicates that successfully fitting frontal identities does not necessarily produce a representation that

remains reliable under substantial viewpoint changes. In contrast, Model B maintained high recognition performance across the pose range represented during training while using the same architecture and training-image budget as Model A. Therefore, the observed improvement is primarily associated with the diversity of pose information provided during training rather than increased model capacity or additional training samples.

Importantly, the benefit of pose-diverse training extended beyond the yaw angles explicitly included in the training set. Model B achieved approximately 95% accuracy on the unseen intermediate poses of $\pm 15^\circ$ and $\pm 45^\circ$. This suggests that the learned representation was able to generalize between the observed pose conditions rather than simply memorizing independent pose-specific appearances. The embedding analysis supports this interpretation: Model B consistently maintained substantially higher same-identity cosine similarity to the frontal representation than Model A across non-frontal views.

The embedding results also revealed a notable relationship between the training-pose distribution and representation stability. Similarity was particularly high at 30° and 60°, which were directly represented during Model B training, while slightly lower values were observed at the intermediate 15° and 45° poses. This pattern suggests that the observed training poses act as representation anchors, while the network interpolates between them for intermediate viewpoints.

Nevertheless, pose diversity did not eliminate pose sensitivity. Performance decreased substantially at 75° and 90°, beyond the maximum 60° yaw used during training. At these extreme views, self-occlusion and the loss of visible facial structure reduce the identity information available to the CNN. The corresponding decrease in embedding similarity indicates that this degradation is associated not only with classifier errors but also with increased pose-induced representation shift. Therefore, the evidence supports both hypotheses, while also showing that training-pose diversity alone remains insufficient for robust recognition under extreme profile views.

8. LIMITATIONS AND RESPONSIBLE USE

This study has several limitations. First, the experiments were conducted using a **single controlled face dataset**. Multi-PIE provides systematic pose and illumination variation, but its acquisition conditions differ substantially from unconstrained real-world imagery such as surveillance video, mobile-camera images, or forensic evidence. Therefore, the reported performance should not be interpreted as expected performance in operational environments.

Second, the task was formulated as **closed-set identification**. The identities present in the test set were also represented during training, although the test images used

held-out illumination conditions and pose variations. The experiment therefore evaluates cross-pose generalization for known identities rather than recognition of previously unseen individuals.

Third, although pose-diverse training substantially improved recognition, it remained sensitive to **extreme unseen poses**. Model B maintained strong performance within and between the observed training poses, but accuracy decreased considerably at 75° and 90°. This suggests that exposure to moderate pose diversity alone cannot fully compensate for the self-occlusion and loss of visible facial evidence that occur in near-profile views.

Finally, the present study does not evaluate demographic fairness, cross-dataset generalization, image degradation, or other real-world nuisance factors such as blur, compression, occlusion, or low resolution. Face-recognition predictions should therefore not be treated as definitive identity evidence. In forensic or other high-stakes applications, model outputs should be interpreted alongside independent evidence and human review rather than used as a standalone decision mechanism.

9. CONCLUSION

This study examined whether pose-diverse training improves cross-pose face recognition and reduces pose-induced representation shift compared with frontal-only training. Under the same CNN architecture, training-image budget, and evaluation protocol, pose-diverse training substantially improved recognition performance across non-frontal views. Across three random seeds, Model B achieved $86.45 \pm 0.51\%$ overall accuracy compared with $19.17 \pm 1.33\%$ for the frontal-only baseline. The pose-diverse model also preserved substantially higher same-identity embedding similarity to frontal representations and generalized well to unseen intermediate yaw angles.

However, recognition and embedding similarity both declined at extreme unseen poses beyond the training range, particularly at 75° and 90°. Overall, the evidence supports both hypotheses: training-pose diversity improves cross-pose recognition and representation stability, but it does not fully eliminate the effects of extreme head pose.

10. REFERENCES

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APPENDIX

A. Additional Training and Performance Results

This section provides supplementary training diagnostics and cross-pose performance analyses that were omitted from the main report for space considerations. The learning curves shown in Figs. A1 and A2 correspond to the representative seed-42 run, whereas Fig. A3 summarizes performance variation across all three training seeds.

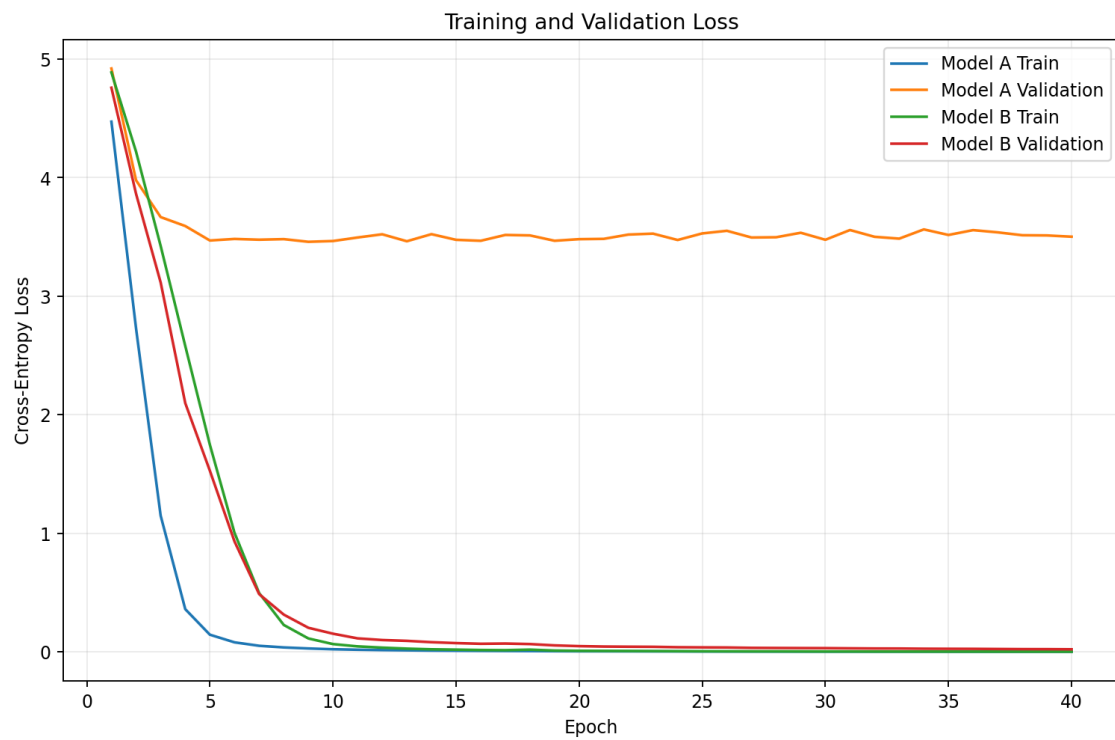


Fig. A1. Representative training and validation loss curves for Models A and B using seed 42.

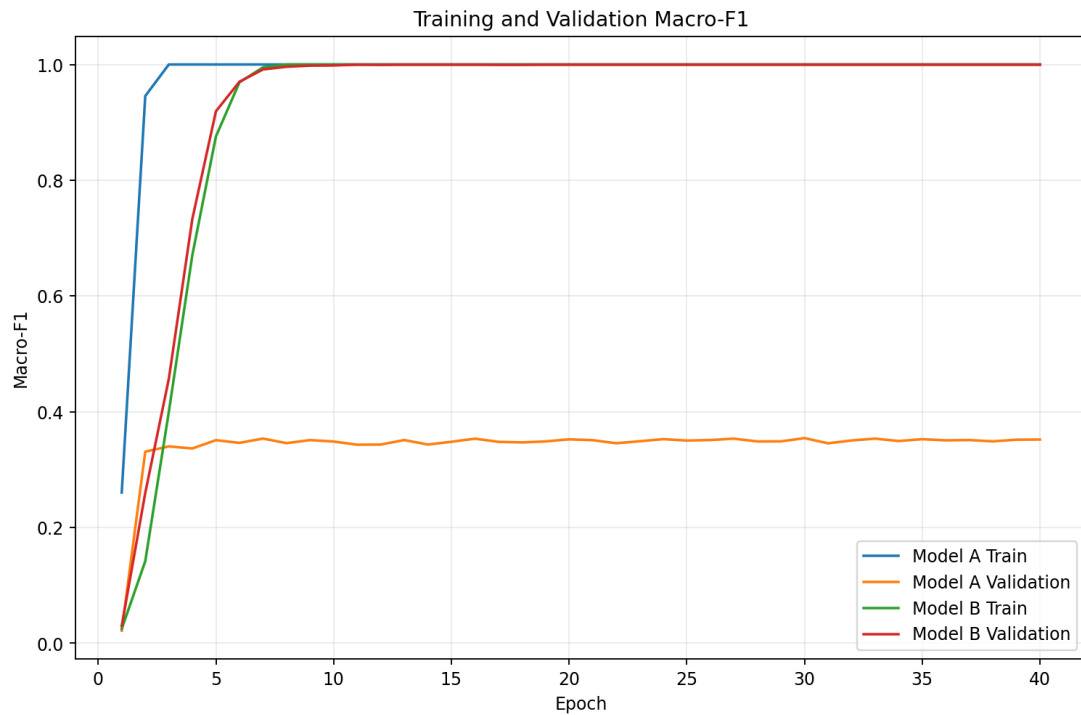


Fig. A2. Representative training and validation Macro-F1 curves for Models A and B using seed 42.

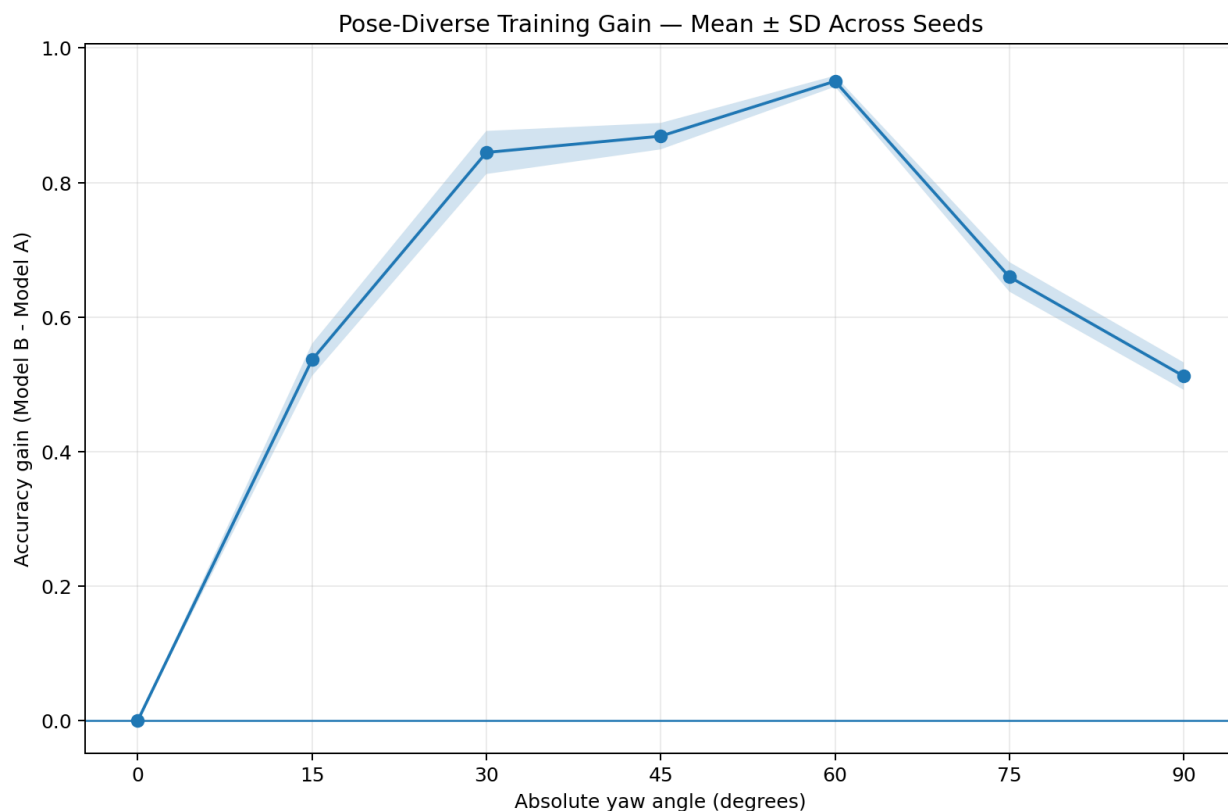


Fig. A3. Accuracy improvement of the pose-diverse Model B over the frontal-only Model A across absolute yaw angles. The line represents the mean across three random seeds, and the shaded region indicates ± 1 standard deviation.

B. Additional Error Analysis

Additional cross-seed error analyses were conducted to examine systematic identity confusions and failures that persisted across repeated training runs. Across all three seeds, 4,258 test images were consistently misclassified by Model A but correctly classified by Model B. In comparison, 556 images were consistently misclassified by both models, while Model B produced 570 persistent failures across all three runs. The persistent errors of Model B were strongly concentrated at extreme yaw angles.

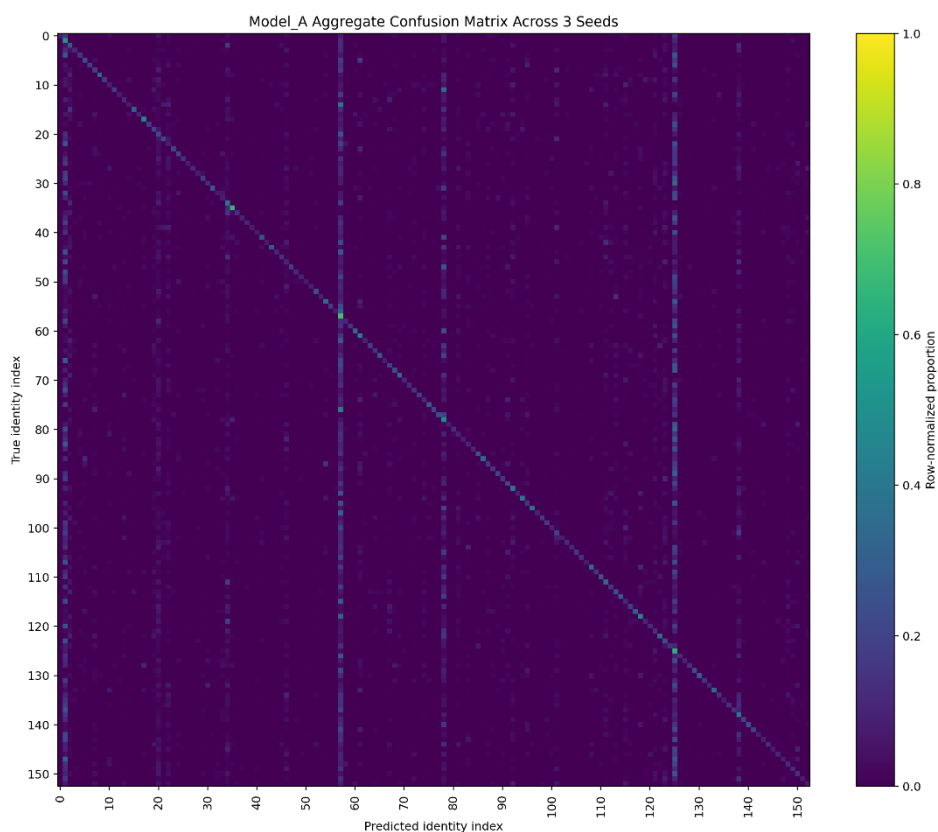


Fig. A4. Aggregate row-normalized confusion matrix for the frontal-only Model A across the three random-seed runs.

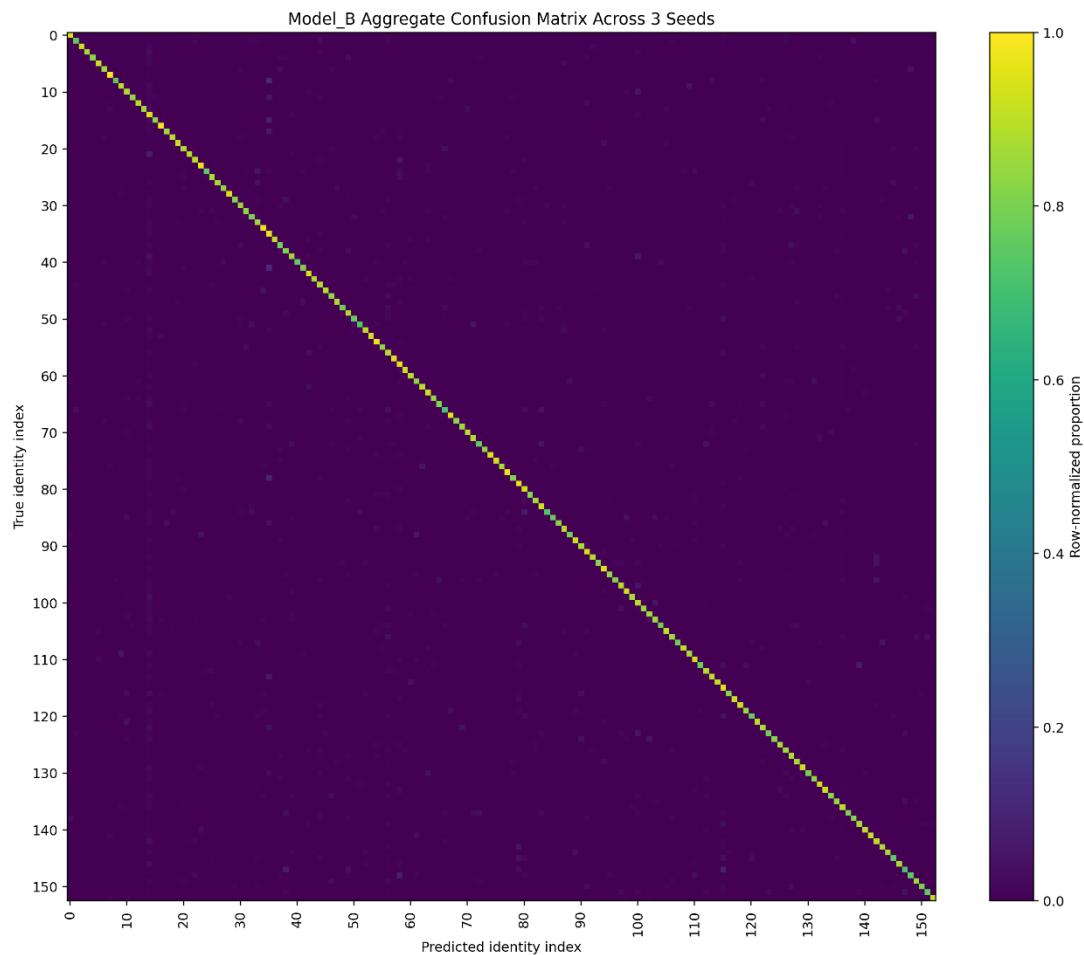


Fig. A5. Aggregate row-normalized confusion matrix for the pose-diverse Model B across the three random-seed runs.

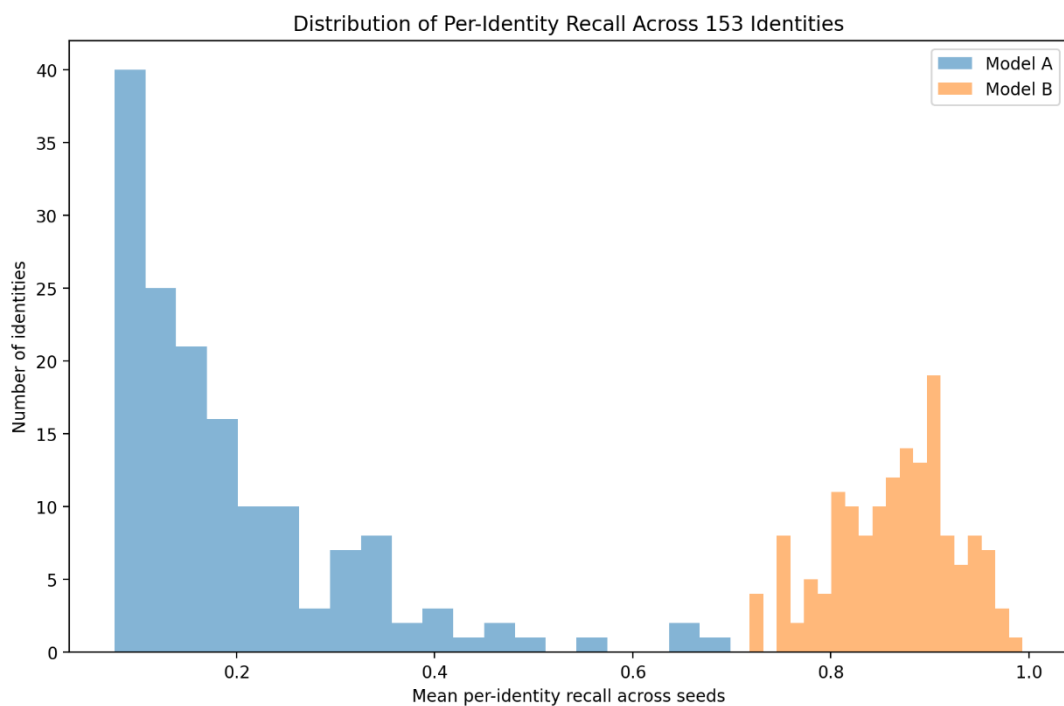


Fig. A6. Distribution of mean per-identity recall across the 153 identities for Models A and B.

The recall distribution further shows that the advantage of Model B was not limited to a small subset of identities. Model B generally maintained substantially higher identity-level recall, whereas Model A exhibited consistently lower recall across the majority of identities.

TABLE A1. PERSISTENT MODEL B FAILURES ACROSS ALL THREE SEEDS

Absolute yaw Persistent failures

0°	0
15°	13
30°	0
45°	18
60°	0
75°	257
90°	282

Persistent failures: both models wrong in all 3 seeds



person002 | yaw -90
A correct seeds: 0/3
B correct seeds: 0/3



person002 | yaw -90
A correct seeds: 0/3
B correct seeds: 0/3



person002 | yaw -90
A correct seeds: 0/3
B correct seeds: 0/3



person005 | yaw -90
A correct seeds: 0/3
B correct seeds: 0/3



person005 | yaw -90
A correct seeds: 0/3
B correct seeds: 0/3



person005 | yaw -90
A correct seeds: 0/3
B correct seeds: 0/3



person005 | yaw +90
A correct seeds: 0/3
B correct seeds: 0/3



person005 | yaw +90
A correct seeds: 0/3
B correct seeds: 0/3



person005 | yaw +90
A correct seeds: 0/3
B correct seeds: 0/3



person006 | yaw +90
A correct seeds: 0/3
B correct seeds: 0/3



person006 | yaw +90
A correct seeds: 0/3
B correct seeds: 0/3



person007 | yaw -90
A correct seeds: 0/3
B correct seeds: 0/3



person007 | yaw -90
A correct seeds: 0/3
B correct seeds: 0/3



person007 | yaw -90
A correct seeds: 0/3
B correct seeds: 0/3



person007 | yaw +90
A correct seeds: 0/3
B correct seeds: 0/3



person009 | yaw -90
A correct seeds: 0/3
B correct seeds: 0/3

Fig. A7. Representative test images that were persistently misclassified by both models across all three random-seed runs.

C. Experimental Configuration

The following tables summarize the fixed data partition and common training configuration used throughout the experiments. The dataset split was generated once and remained unchanged across all model comparisons and repeated-seed runs.

TABLE A2. FIXED ILLUMINATION-BASED DATA SPLIT

Split	Illumination IDs	Conditions
Training	1, 4, 6, 7, 8, 10, 11, 13, 15, 16, 17, 20	12
Validation	3, 5, 12, 19	4
Test	2, 9, 14, 18	4

The illumination split was generated using seed 42 and was fixed for all experiments. The repeated training seeds (42, 123, and 2026) affected model initialization, data shuffling, and augmentation randomness only; they did not modify the train, validation, or test partitions.

TABLE A3. COMMON TRAINING CONFIGURATION

Parameter	Setting
Input size	$128 \times 128 \times 1$
Number of identities	153
Loss function	Cross-entropy
Optimizer	Adam
Learning rate	3×10^{-4}
Weight decay	1×10^{-5}
Batch size	32
Training epochs	40
Gradient clipping	Maximum norm = 5.0
Embedding dimension	128
Training seeds	42, 123, 2026
Model selection	Highest validation Macro-F1
Tie-breaking rule	Lower validation loss
Augmentation	Brightness $\pm 10\%$, contrast $\pm 10\%$
Geometric augmentation	None

TABLE A4. TRAINING-POSE ALLOCATION PER IDENTITY

Model	Training yaw allocation	Images / identity	Total images
Model A	0°: 12	12	1,836
Model B	0°: 4, − 30°: 2, + 30°: 2, − 60°: 2, + 60°: 2	12	1,836

Both models therefore used exactly the same number of original training images. Training-pose distribution was the primary experimental variable.

D. Additional Dataset Visualization

Fig. A8 provides an enlarged visualization of the pose variation represented in the experimental dataset. The examples complement the compact pose illustration shown in Fig. 1 of the main report and allow the facial appearance changes across yaw angles to be inspected more clearly.

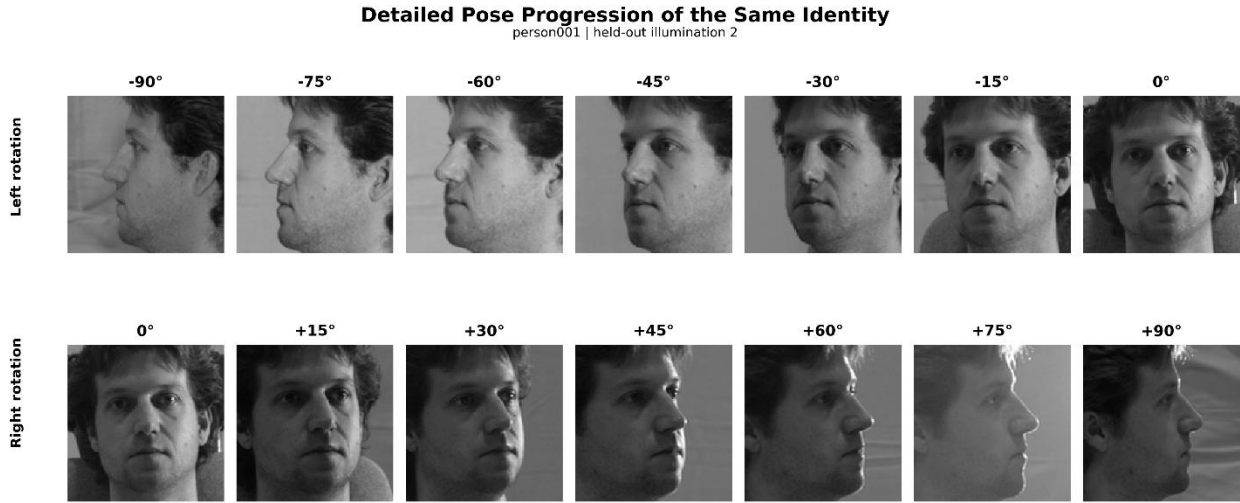


Fig. A8. Enlarged examples of the same identity across left and right yaw rotations in the Multi-PIE subset. Increasing yaw produces progressively stronger appearance changes and self-occlusion.